

Global assessment of the carbon-water tradeoff of dry cooling for thermal power generation

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Abstract. Water scarcity and climate change are dual challenges that could potentially threaten energy security. Yet, integrated water-carbon management frameworks coupling diverse water- and carbon-mitigation technologies at high spatial heterogeneity are largely underdeveloped. Here, we build a global unit-level framework to investigate the CO₂ emission and energy penalty due to the deployment of dry cooling—a critical water mitigation strategy—together with alternative water sourcing and carbon capture and storage (CCS) under climate scenarios. We find that CO₂ emission and energy penalty for dry cooling units are location and climate specific (e.g., 1-15% of power output), often demonstrating significantly faster efficiency losses than rising temperature, especially under the high climate change scenario. Despite energy and CO₂ penalties associated with alternative water treatment and CCS utilization, increasing wastewater and brine water accessibility provide potential alternatives to dry cooling for water scarcity alleviation, whereas CO₂ storage can help to mitigate dry cooling-associated CO₂ emission tradeoffs when alternative water supply is insufficient. By demonstrating an integrative planning framework, our study highlights the importance of integrated power sector planning under interconnected dual water-carbon challenges.

Thermal electric power generation uses substantial amounts of freshwater primarily for cooling, amounting to approximately 40-50% of total water withdrawal in the U.S. and 40% in Europe¹⁻³. Meanwhile, power generation accounts for approximately 36% of energy-related carbon dioxide (CO₂) emissions across advanced economies in 2019, placing the power sector among the world's largest CO₂ emitters⁴. Consequently, the power sector has a high dependence on freshwater resources and demonstrates intrinsic water-carbon interconnections, which thereby have critical implications on reliable electricity output and energy security, particularly under climate change^{5,6}.

With increasing recognition of the inherent water-carbon interconnections in electricity generation, growing studies have been calling for integrated water and energy systems planning^{3,7}. Previous work mostly fix on the water-energy nexus via evaluating the climate impacts on usable electricity generation capacity and associated power sector vulnerability due to water quantity and water temperature changes⁸⁻¹³. Recent efforts unraveling the power sector water-carbon linkages often center on those more popular fuel switch strategies (e.g., switching from coal to natural gas^{14,15} and to low/no-water consuming renewables^{16,17}). In comparison, the underlying water-carbon interactions for critical water mitigation and retrofitting technologies in power sector water management have been relatively under-evaluated, particularly for the most progressive water-mitigating yet energy-consuming dry cooling techniques, which are often analyzed within limited geographic locations^{6,10,18}, at aggregated resolutions¹⁸, or treated with relatively simplified assumptions¹⁹. Meanwhile, increasing studies are developing global consistent high-resolution unconventional water resource datasets²⁰⁻²², which advance the evaluation of the potential role of alternative water sources (e.g., wastewater and brine water) in alleviating water stress^{6,23,24}, though so far often not focus on the power sector or are constrained in limited geographic regions⁶.

Facing increasing water scarcity concerns, dry cooling has been and may continually be promoted as an emerging freshwater mitigation technique in some major economies in the next few decades along with renewable energy transition^{5,18}. In spite of its water saving characteristics, dry cooling is disadvantaged by higher costs and larger land footprints. Most notably, it suffers from reduced efficiencies due to two key factors: the use of fans to increase airflow required for steam condensation results in an additional parasitic load thus reducing the useful electrical output of the plant; and, higher temperature results in a higher “backpressure” (e.g., a negative pressure that has to be overcome for the steam flow to continue), which reduces the temperature gradient across the steam turbine thus also reduces the electrical output. Combined, these factors increase fuel consumption and CO₂ emissions per unit of electricity produced^{2,3,18}. Given the intrinsic water-carbon interlinkages, dry cooling could provide one of the best examples to illustrate the role of integrated power planning, especially in the context of alternative water sourcing and carbon mitigation technologies in resolving the dual challenges of water scarcity and climate change. However, the geospatial variations, temporal trends, and potential technology substitutes for dry cooling associated water-carbon interactions have not been systematically evaluated at the global scale, thereby inevitably hindering an integrated and proactive power sector planning.

Here, we design a global unit-level integrated framework to systematically investigate dry cooling associated water-carbon interactions by coupling alternative water sourcing and carbon mitigating technologies. Integrating location-specific generation units (e.g., fuel type and installed capacity), local meteorology (e.g., air temperature and relative humidity), together with water properties (e.g., water temperature and runoff), we first characterize the geospatial patterns of dry cooling associated water-carbon interconnections. We then identify global hotspots that could be particularly vulnerable for dry cooling deployment due to increasing efficiency losses under climate warming scenarios. Integrating information on alternative water sourcing and CO₂ mitigation techniques, we further explore utilizing alternative sources of cooling water and CO₂ storage capacity to better tackle the dual challenges of water scarcity and climate change via establishing an integrated cooling technology-alternative water sourcing-CO₂ storage framework.

Age-based water-carbon interactions for dry cooling units

Fig. 1 demonstrates the age structure of global fossil (e.g., coal, natural gas, and oil) and non-fossil (e.g., biomass, solar and nuclear) dry cooling units in 2015 broken down by major countries/regions. We observe three dominant global features: First, fossil fleets dominate dry cooling generation units. Among ~177 GW global major dry cooling units evaluated here in 2015, coal accounted for ~86% (152 GW) of total operational capacity, followed by natural gas (12%, 21 GW) and oil (1.2%, 2.1 GW) fleets (Supplementary Fig. 1). Second, dry cooling units are younger than generation units equipped with other cooling technologies, indicating a much more recent development and deployment. In 2015, approximately 51% (91 GW) and 82% (146 GW) of dry cooling generation capacity were less than five and ten years old, respectively. In comparison, roughly 25% (44%) and 10% (16%) of recirculating and once-through-freshwater generation units are less than 5 (10) years old²⁵, respectively. Once-through has historically been the dominant cooling technology due to its relatively simple design and low costs, and the fact that earlier power plants are often built close to abundant water resources^{3,26}. However, with increasing concerns on local water scarcity, thermal pollution, ecological disruption and therein growing regulatory pressure²⁷, recirculating and dry cooling are now favored both in the design of new plants and major retrofits²⁵. Until more recently, dry cooling generation units have historically only been used in a few countries. In particular, China alone contributes ~73% of global total dry cooling capacity, almost all of which are coal units (Supplementary Fig. 1). South Africa, U.S., Iran, and India also have notable dry cooling capacity; although other than in South Africa and India, most are non-coal units (e.g., a large chunk of natural gas dry cooling fleets in the U.S.). Consequently, dry cooling-associated water-carbon tradeoffs are most evident in China, South Africa, and India, where dry cooling units primarily burn high carbon-emitting coal.

Dry cooling-induced freshwater mitigation is achieved at the expense of increased CO₂ emissions. As illustrated in Fig. 1 a and d and Supplementary Fig. 2, among dry cooling fossil fleets, increased CO₂ emissions and avoided water withdrawal across age groups are roughly in proportion to their respective generation capacity, though less so for non-fossil fleets. Globally, dry cooling techniques applied to major fuel types avoid approximately 75 and 0.4 billion cubic

149 meters (bcm) of water withdrawal and water consumption, respectively, equaling to roughly
150 complete water withdrawal and 84% of water consumption mitigation in comparison to once-
151 through cooling. These water savings are achieved at the expense of approximately 39 million
152 tonnes (Mt) of increased CO₂ emissions, that is around 5% extra emissions when compared with
153 once-through cooling. In comparison, switching from recirculating to dry cooling can save 1.3
154 (95%) and 1.9 bcm (93%) of water withdrawal and water consumption, respectively, at the expense
155 of approximately 34 Mt (6%) increases in CO₂ emissions. This demonstrates apparent dry cooling-
156 induced water-carbon tradeoffs, particularly for notable water withdrawal mitigation.

158 **Geospatial variations in water-carbon interactions**

159 The spatial pattern of dry cooling deployment has large implications on local water scarcity,
160 location-specific efficiency loss, CO₂ emissions, and hence the water-carbon interactions, in
161 comparison to counterfactual cases of once-through (Fig. 2) and recirculating (Supplementary Fig.
162 3) cooling utilization. Across the globe, unit-level avoided water withdrawal ranges from ~0.4 to
163 800 million m³/yr in comparison to once-through cooling and ~0.01 to 15 million m³/yr in
164 comparison to recirculating cooling, respectively. The largest unit-level avoided water withdrawal
165 is primarily centered in northern China, western U.S., and southern Africa. Cross-unit variations
166 in avoided water withdrawal for the same counterfactual cooling techniques are mainly due to
167 differences in unit-level electricity generation and in water withdrawal coefficients owing to
168 different fuel types and location-specific meteorology. Retrofitting from once-through to dry
169 cooling, on average saves roughly 50 times more water withdrawal, than switching from
170 recirculating to dry cooling (Fig. 2a and Supplementary Fig. 3a), primarily due to orders of
171 magnitude larger water withdrawal coefficients of once-through fleets than other cooling
172 techniques¹⁹. The spatial pattern of avoided water consumption is largely resembling that of
173 avoided water withdrawal, yet water consumption savings are larger when retrofitting from
174 recirculating cooling to dry cooling than switching from once-through to dry cooling, but their
175 differences are much smaller (~2.5times) than water withdrawal (Supplementary Fig. 4).

176 Increases in efficiency loss and CO₂ emissions demonstrate notably different spatial patterns to
177 the patterns of avoided water withdrawal. Overall, dry cooling efficiency losses are roughly
178 1.4–15% and 1.3–13% (as a percent of electricity output) higher than once-through and
179 recirculating cooling, with the hotspots of high-efficiency-loss concentrating around India,
180 Southeast Asia, and southern parts of Africa. Units with high levels of increased CO₂ emissions
181 are primarily centered around India and northern China (i.e., 0–550 kilo tonnes in comparison to
182 once-through cooling and 0–510 kilo tonnes to recirculating cooling), where dry cooling units
183 mostly use coal as input fuel, and have relatively high efficiency loss and/or large generation
184 capacity. Geospatial patterns of dry cooling-associated efficiency loss increases do not resemble
185 that of CO₂ emission increases as generation units' installed capacity and fuel-specific carbon
186 emission factors also vary by region. Despite notable dry cooling-induced water savings, the
187 associated CO₂ emissions can be substantial. In particular, due to high ambient temperature

exposure and dominating coal fleets in India, it is among the global hotspots of high carbon penalties (i.e., increased CO₂ emissions) with per unit avoided water withdrawal (Supplementary Fig. 5).

Tackling dry cooling fleets with high-efficiency-loss

We observe substantial variations in dry cooling units' associated efficiency losses across fuel types, geographic regions and seasons predominantly driven by exposure to high ambient temperature (Fig. 3 and Supplementary Fig. 6), which could provide potential targeted mitigation opportunities via utilizing alternative water sources or carbon mitigation technologies.

Primarily driven by ambient temperature (Supplementary Fig. 6a), oil fleets have the largest fraction of units exposed to relatively high efficiency loss (i.e., efficiency loss as a fraction of expected power generation $\geq 7.5\%$), followed by biomass and coal units. High-efficiency-loss biomass fleets contribute roughly 30% of total generation capacity and avoided water withdrawal, and almost 45-50% of increased CO₂ emissions. To a lesser degree, a similar pattern is observed for coal fleets. Due to the dominating coal share within dry cooling fleets (~86%), it makes up the largest total amount of high-efficiency-loss dry cooling units (~24 GW out of 29 GW). In comparison, natural gas fleets have a much larger share of low-efficiency-loss units (i.e., efficiency loss as a fraction of expected power generation $< 2.5\%$), with roughly 8% of fleets exposed to high efficiency loss, which contributes ~10% of avoided water withdrawal and ~15% of increased CO₂ emissions. A large share of high-efficiency-loss oil, biomass and coal dry cooling fleets raises critical concerns regarding water-carbon tradeoffs. Although CO₂ emissions from biomass combustion are considered close to carbon neutral, high-efficiency-loss biomass and fossil fleets, potentially provide targeted carbon mitigation opportunities via carbon capture and storage (CCS) [28-29](#).

Likewise, across geographical regions, Africa has the highest proportion of units exposed to high efficiency loss due to a large portion of units exposed to high temperature (Supplementary Fig. 6b), with roughly 80% (~9.7 GW) of such operating capacity. In comparison, a much smaller share of dry cooling generation capacity in Asia (~12%), North America (~4%), and Europe (~1%) are high-efficiency-loss units. Nevertheless, with the largest quantity of dry cooling generation capacity (~140 GW), Asia alone (~17 GW) contributes ~55% of global total high-efficiency-loss units, highlighting regions exposed to the largest resulting energy and carbon penalties.

Additionally, as over 90% of dry cooling fleets are concentrated in the northern hemisphere, we observe a strikingly larger fraction of high-efficiency-loss dry cooling fleets in northern hemisphere summer than in the three other seasons due to higher summer temperature (Supplementary Fig. 6c). For example, in June-July-August, high-efficiency-loss fleets make up ~85% of seasonal total installed dry cooling capacity, compared with merely 8–13% of fleets are exposed to high efficiency loss in other three seasons.

Increasing efficiency loss under a warming climate

Increasing ambient temperature under a warming climate generally leads to higher backpressure and consequently nonlinearly increasing higher efficiency loss for dry cooling units, which could thereby threaten a reliable electricity output (Supplementary Table 1 and Extended Fig. 1). On global average, other than RCP2.6, we observe a consistent increase in both ambient temperature and efficiency loss over the years, with the most striking increases observed under RCP8.5. Global weighted average efficiency loss of dry cooling units increases from ~6.4% in 2020-2029 to 6.7% and 7.1% in 2050-2059 and to 7.4% and 8.4% in 2090-2099 under RCP6.0 and RCP8.5, respectively, indicating a proportional electricity generation loss from dry cooling fleets. Notably, dry cooling fleets' efficiency loss generally demonstrates faster increases than ambient temperature, particularly under RCP8.5 (Fig. 4, Extended Fig. 1, and Supplementary Fig. 7). Fig. 4 (d-f) illustrates the spatial pattern of the relative scaling factors of efficiency loss to ambient temperature under each RCP scenario, which further highlights a faster-than-temperature increase in efficiency losses under a warming climate, particularly under RCP8.5. Similar trends are also observed for once-through and recirculating cooling units, with the most prominent efficiency losses under the high climate scenario (Supplementary Figs. 8-11).

Integrated power sector planning under climate change

Under expected intensifying droughts^{30,31}, increasing water demand³², and stricter regulations¹⁸, dry cooling techniques may play a more important role before renewables could largely substitute fossil fleets³. Here hence we explore a hypothetical case that, among global thermal units exposed to high water scarcity (i.e., water scarcity index: $WSI > 0.4$ ^{32,33}) that aren't currently equipped with dry cooling techniques (i.e., once-through and recirculating cooling, Supplementary Fig. 12), what would their efficiency loss be if they were retrofitted to dry cooling to avoid further water scarcity. Fig. 5a presents units-specific efficiency loss if dry cooling were employed, showing generation fleets with high efficiency loss and sizeable capacity are primarily concentrated in India, eastern China, western Europe, Southeastern Asia, and southern U.S., signaling the necessity to integrate alternative technologies into power sector planning.

Thereby, we further explore the potential role of alternative water sourcing (e.g., wastewater and brine water with commercial-prospects (C-brine), see *Methods*) and CO₂ storage in resolving dry cooling associated water-carbon tradeoffs if generation units would suffer from relatively high efficiency loss when switching to dry cooling (blue points in Fig. 5a). Fig. 5 (b-e) shows the ratio of location-specific total cooling water withdrawal to alternative water availability. Increasing the reusable proportion-share of wastewater reused (WWr) to total produced wastewater (WWp)-plays a most important role in providing sufficient alternative water, while increasing the collection distance for wastewater and brine water can also significantly increase alternative water availability. If all WWp could be reused with a collection distance of 25 km, the proportion of thermal generation capacity whose cooling water demand could be fully satisfied (i.e., satisfiable portion) via wastewater and brine water would increase from 25% (WWr+C-brine) to 85% (WWp+C-brine). In comparison, increasing alternative water (WWr+C-brine) collection distance from 10 to 50 km, this share could increase from 15% to 42%.

As currently-identified commercial ready CO₂ storage and the associated brine water (C-brine) is relatively small, the satisfiable portion is predominately provided by wastewater (Supplementary Fig. 13a-b). Therefore, the brine water extraction rates from commercial-ready carbon storage candidate sites have negligible impacts on the satisfiable portion. Nevertheless, global total geologic CO₂ storage capacity and the corresponding brine water availability identified by the United States Geological Survey (USGS) are significantly larger than those identified with relatively high commercial readiness³⁴ (Supplementary Figs. 14-15). Thus, if global total geologic CO₂ storage capacity can be developed, the associated extractable brine water together with reusable wastewater (WWR+USGS-brine, Fig. 5e) can meet roughly 63% of thermal water demand under 25 km collection distance, and could further increase to 73% when expanding the collection distance to 50 km (Supplementary Fig. 16a). As brine water availability associated with candidate USGS carbon storage sites (USGS-brine) is much larger, it can surpass WWR in meeting thermal generation water demand, although still smaller than WWp (Supplementary Fig. 13c and 16b). For generation units whose nearby alternative water is insufficient in meeting their water withdrawal (e.g., magenta points in Fig. 5e), we thereby still have to rely on dry cooling for water scarcity alleviation, and hence further turn to carbon storage to resolve dry cooling-induced CO₂ emission increases. Roughly 19.5 GW out of 141 GW (magenta points in Fig. 5e), that is ~14% of thermal generation capacity have access to nearby USGS geological CO₂ storage (25 km accessibility distance). Therefore, other strategies (e.g., switching to fuel types requiring no/low water uses, wind turbine and solar PV; inter-basin water transfer) may be needed for those remaining units without close access to CO₂ storage, particularly in India, eastern China, and Thailand (Fig. 5f).

Therefore, challenges exist in largely relying on currently reusable wastewater and brine water (e.g., particularly C-brine) to substitute dry cooling in alleviating water scarcity or relying on potential carbon storage to address dry cooling-induced CO₂ penalties, especially considering there will be increasing energy consumption and/or water uses accompanying alternative water treatment and CCS (Supplementary Notes, Supplementary Figs. 17-18, Supplementary Tables 2-3)¹⁸. That said, thermal units are mostly dominated by net water savings and CO₂ emission reductions (Supplementary Figs. 17-19). Therefore, increasing reusable wastewater and more progressive brine water extraction (e.g., USGS-brine) to substitute for dry cooling, or alternatively, dry cooling coupled with CO₂ storage can still potentially provide important opportunities to partly resolve power sector associated water-carbon tradeoffs.

Discussion

Cooling water availability directly affects thermoelectric power generation capacity and electricity system reliability^{8,9}. Climate-induced changing water availability together with intensifying cross-sector water use competition^{9,10,32,35} are expected to pose increasing threats to global power system that has a strong dependence on freshwater resources^{3,6} and contributes substantial CO₂ emissions which can further aggravate climate change³⁶. Therefore, an integrated power sector design to address the underlying water-carbon nexus is essential. In particular, dry cooling serves as a great candidate to illustrate integrated power planning via coupling water use reduction

(i.e., dry cooling), water supply expansion (i.e., alternative water sourcing), and CO₂ mitigation (i.e., carbon storage) technologies due to its intrinsic water-carbon tradeoffs.

Despite dry cooling consistently reduces water use at the expense of increased energy consumption and/or CO₂ emissions, it demonstrates notable heterogeneities regarding the magnitude of dry cooling associated energy penalties and efficiency losses across fuel types, geographic regions, and four seasons. Such variations are primarily driven by dry cooling fleets' exposure to ambient temperature, with higher temperature exposure inducing higher backpressure, which thereby generally requires extra energy to overcome the backpressure, and consequently causes higher energy penalty and lower overall energy efficiency (Supplementary Fig. 6). Such characteristics could thereby pose critical challenges to dry cooling units for freshwater mitigation and energy security: Dry cooling is mostly needed in drought and arid regions that are exposed to high water scarcity; with an expected intensifying water scarcity under a warming climate, there may be an increasing reliance on dry cooling techniques. However, owing to significantly faster increasing rates in efficiency loss than in ambient temperature for dry cooling fleets, dry cooling associated water-carbon tradeoffs could be further intensified under climate change. Global weighted average efficiency loss and hence associated electricity losses of dry cooling units can reach 6.7% and 7.1% in 2050-2059 and 7.4% and 8.4% in 2090-2099 under RCP6.0 and RCP8.5, respectively.

This is particularly concerning under the warmest scenario, especially in regions such as India, where large amounts of generation fleets are already exposed to high water scarcity and where both thermal generation capacity are expected to further increase, yet the associated efficiency losses and corresponding electricity generation loss of local dry cooling fleets can exceed 15% in the middle of the century under a warming climate. Consequently, a warming climate could induce higher dependence on dry cooling yet simultaneously make it less energy efficient and pose potential threats to local energy supply, especially when dry cooling is largely employed and under notably high temperature. In that vein, additional water- and carbon-mitigation strategies are also needed to better resolve the dual challenges of water scarcity and climate change.

Unconventional water supply (e.g., wastewater and brine water) can potentially compensate for freshwater inadequacy to alleviate water scarcity. With a collection distance of 25 km, reusable wastewater (WWr), total produced wastewater (WWp), commercial-ready brine water (C-brine), and total geological brine water (USGS-brine) can individually meet 24.3%, 85.1%, 0.9%, and 56.3% of thermal water demand, or 39.4%, 96.6%, 3.2%, and 61.1% under 50km, demonstrating both total produced wastewater and USGS-brine water can largely substitute dry cooling in tackling water scarcity, highlighting the importance of increasing the reusable proportion of wastewater, expanding alternative water collection distance, and progressive brine water extraction. When nearby alternative water is insufficient to substitute dry cooling in mitigating water scarcity, USGS carbon storage can help to tackle dry cooling-induced CO₂ emission penalties. In spite of extra energy and/or water demand for alternative water treatment and CCS employment, alternative water sources and carbon storage can generally provide important

opportunities to resolve the water-carbon tradeoffs in the power sector (Supplementary Figs. 17-19). Such trends are generally consistent under a changing climate (Supplementary Figs. 20-21).

Limitations and uncertainties apply to this study. First, we focus on unit-level cooling technology deployment together with grid-level alternative water sourcing and site-specific carbon storage information to establish an integrated power planning framework. Yet, factors such as inter-basin water transfer, groundwater pumping, and virtual water transfer are not considered in our current unit-level framework, as such high-resolution data are still largely lacking across the globe. When future fine-resolution data becomes available, we can incorporate such information into our unit-level framework for improvement. Second, our estimated brine water is conservative as we only consider brine extraction from candidate CO₂ storage sites. This is because brine extraction from CO₂ storage sites can serve as a pressure management practice to prevent potential hazards such as land subsidence via CO₂ injection, and significantly reduce brine extraction costs as it may be partially covered by geological carbon storage. Third, our results can potentially be affected by uncertainties in climate internal variability, we thereby rely on multi-model ensemble means in combination with the model spreads to represent the most likely results together with the uncertainty ranges. Supplementary Figs. 22-23 show the standard deviation and coefficient of variation on unit-level efficiency losses based on model-specific results, indicating our results are generally robust across climate models. Fourth, here we only consider dry cooling generation units directly recorded in the world electric power plants database, which may have underestimated total dry cooling capacity. However, the spatial and temporal patterns of dry cooling associated water-carbon tradeoffs are consistent, and the integrated framework can be extended to other dry cooling units relatively straightforward.

As our integrated framework also identifies the global hotspots where neither dry cooling nor alternative water sourcing or carbon storage are capable of easily resolving the water-carbon tradeoffs from thermal generation (e.g., India and east China in Fig. 5f), our study thereby also indicates where switching to no- or low-water consuming renewables (e.g., wind and PV) are most needed from the water scarcity alleviation perspective, indicating potential water-oriented power sector transition pathways along with the worldwide carbon neutrality pursuit. Because of the intrinsic water-carbon interlinkages, it is important to also factor into water concerns in designing worldwide power sector decarbonization pathways, particularly for regions such as India and Africa that may simultaneously suffer from intensifying water scarcity and increasing energy demand.

By systematically unraveling unit-level water-carbon interlinkages associated with dry cooling in the context of alternative water sourcing and carbon storage, our study provides a global-scale unit-level framework for integrated power sector planning, which is becoming increasingly indispensable under complex challenges facing human populations.

Methods

Unit-level water use and CO₂ emissions. We estimate unit-level water withdrawal, water consumption, and CO₂ emissions for global dry cooling generation units operating in 2015, primarily based on the World Electric Power Plants (WEPP) database (2017 version) (<https://www.spglobal.com>)³⁷, which provides basic power generation units information, such as fuel types, engine types, installed capacity, cooling technology, and administrative-level company information²⁵. We then obtain unit-level geo-coordinates and additional information from Qin et al. (2019)²⁵, which primarily extract information from existing power databases (e.g., CARMA and eGRID), the World Cities information, and Google application programming interfaces (APIs). We estimate unit-level electricity generation by multiplying each unit's installed capacity and its corresponding capacity factor. We obtain fuel-specific CO₂ emission factors from the Intergovernmental Panel on Climate Change (IPCC) reports (<https://www.ipcc-nggip.iges.or.jp/public/2006gl/>) Water withdrawal and water consumption coefficients for different generation units are obtained based on earlier studies^{19,25,38-41}. For each generation unit, we then estimate its CO₂ emissions, water withdrawal, and water consumption based on unit-level electricity generation and corresponding emission factors, water withdrawal, and water consumption coefficients. Note we include CO₂ emission estimation from biomass combustion, which are considered close to be carbon neutral, as it can potentially provide carbon mitigation opportunities via carbon capture and storage (CCS).

Unit-level efficiency loss. We estimate unit-level efficiency loss based on the WEPP database and hydrological model outputs. We first simulate grid-level water temperature using the PCRaster Global Water Balance model version 2 (PCR-GLOBWB 2), a state-of-the-art grid-level global hydrology and water resources model, at 5 arc-minutes spatial resolution^{42,43}, which has five major modules: meteorological forcing, land surface, groundwater, surface water routing, and irrigation and water use⁴². Five global climate models (GCM models: GFDL-ESM2M, HadGEM2-ES, IPSL-CM5A-LR, MIROC-ESM-CHEM, and NorESM1-M) from the Inter-Sectoral Impact Model Inter-comparison (ISI-MIP) Project forced by three different Representative Concentration Pathway emissions scenarios (RCP2.6, RCP6.0, and RCP8.5) in the Coupled Model Intercomparison Project Phase 5 (CMIP5) are used in this study to provide input meteorological variables (e.g., air temperature, air pressure, humidity) at the spatial resolution of 0.5× 0.5 degrees for PCR-GLOBWB 2 model and our estimation of unit-level efficiency loss. As explained in earlier studies^{44,45}, PCR-GLOBWB 2 employs an improved water temperature module compared to the previous energy balance model, which leads to better comparison between water temperature simulation and observation due to improved physical realism (e.g., additional processes to cover ice breaking up and larger water bodies thermal mixing). Refer to Sutanudjaja et al. for more details on PCR-GLOBWB 2⁴². For each GCM and RCP scenario combination, we conduct simulations for both the historical (1996-2005) period and the future climate (2020-2099). 10-yr monthly mean values (e.g., 1996-2005) from each model output under each RCP scenario are used to represent the average historical and future climate conditions.

With the same end-use electricity output, power generation units equipped with dry cooling technology have notable efficiency loss and associated energy and/or carbon penalty (i.e., increased energy consumption and CO₂ emissions) due to higher backpressure and lower efficiency of steam turbines, as well as more energy consumption for fans and pump operations^{10:46}. We focus on the energy penalty in steam turbines and combined cycle power plants (e.g., coal, natural gas, oil, biomass, nuclear, and concentrated solar power). Based on grid-level 10-yr monthly average meteorological inputs and water temperature outputs of the PCR-GLOBWB 2 model and unit-level technology information (e.g., fuel types and installed capacity), we estimate energy penalty for global dry cooling generation units using Equations 1-5. For each dry cooling generation unit, we also estimate its counterfactual efficiency loss assuming once-through or recirculating cooling technology were employed instead (Eq. 6). We then compare the relative efficiency losses, as well as changes in water withdrawal, water consumption, and CO₂ emissions for those dry cooling fleets as in comparison with the hypothetical cases of once-through and recirculating cooling, respectively (Figs. 2 and supplementary Fig. 3).

Steam turbine efficiency loss as a percentage of designed electricity output ($\Delta\eta$) is a function of turbine pressure (p, inches Hg). Following earlier studies⁴⁶, Equations 1-4 are used to estimate turbine backpressure for fossil steam (including biomass, 67% load), nuclear (100% load), combined cycle (67% load), and other generation units (67% load), respectively. As pointed out in earlier studies, nuclear fleets often used as the baseload mostly operating closer to full capacity (represented by the 100% load curve), yet fossil units often operate closer to the 67% load curve⁴⁶. Backpressure here refers to engine-produced exhaust gas pressure to overcome the exhaust system's hydraulic resistance, so as to allow steam flow to continue. Thereby, increasing backpressure associated with wet and dry cooling towers will lead to reduced turbine efficiencies and consequently power generation penalties. T_{as} and T_{in} represent ambient air temperature (Degree F) and condenser inlet temperature (Degree F), respectively⁴⁶. The condenser inlet temperature is estimated as water temperature for once-through cooling generation units, while equals cooling tower outlet temperature for recirculating generation units⁴⁶. Cooling tower performance is largely determined by the difference between wet bulb temperature and the cooling tower outlet temperature, which is referred to as tower approach^{10:46}. In addition to steam turbine efficiency loss, cooling generation units also lose efficiency due to fans and pumping equipment, as summarized below. Therefore, efficiency loss in this study is estimated as a percent of designed electricity output⁴⁶. We further classify dry cooling units into four classes roughly following the 25th (2.5%), 50th (5%), and 75th (7.5%) percentiles of unit-level efficiency loss.

Based on historical unit-level efficiency losses for dry cooling units, we sum up their generation capacity, electricity generation, as well as dry cooling fleets' avoided water withdrawal and increased CO₂ emissions in comparison to water-cooling technologies, by major fuel types, regions, and seasons (Fig. 3). Such targeted opportunities identify fuel types, geographic regions, and seasons when or where efficiency losses are particularly high for dry cooling generation units, which hence require particular attention.

On top of that, we further analyze the changes in unit-level and aggregated efficiency losses for dry cooling fleets under different climate scenarios (Extended Fig.1 and Fig. 4). We categorize the future years into three time periods: early century (2020-2039), mid-century (2040-2059), and late century (2060-2099). While existing dry cooling units are most likely to be retired in early and mid-century, we still provide the efficiency losses for the whole future period up to 2099, as our primary focus is to evaluate the potential efficiency losses evolution for existing or similar dry cooling generation units under different levels of future climate (RCP2.6, RCP6.0, and RCP8.5). In addition, we also evaluate efficiency loss changes for the counterfactual retrofitting cases of dry cooling units to once-through and recirculating cooling (Supplementary Fig. 8).

1) Turbine efficiency loss (as a percent of designed power output)

Combustible steam	$\Delta\eta_{(1)} = 0.0063 * p^2 - 0.004 * p - 0.0062$
Nuclear steam	$\Delta\eta_{(2)} = -0.0006 * p^3 + 0.0099 * p^2 - 0.0208 * p + 0.0111$
Non-nuclear steam non-combustible	$\Delta\eta_{(3)} = -0.0013 * p^3 + 0.0169 * p^2 - 0.0286 * p + 0.0098$
Combined cycle	$\Delta\eta_{(4)} = -0.0004 * p^3 + 0.0082 * p^2 - 0.016 * p + 0.0033$

2) Exhaust back pressure (inches Hg)

Dry	$p = 1.031 * \exp(0.019 * T_{as})$	(5)
Once-through/ Recirculating	$p = 0.4591 * \exp(0.0213 * T_{in})$	(6)

3) Fans and pumping energy use (as a percent of designed power output)

	Combustible steam	Non-combustible steam	Combined cycle
Dry	2.43%	0.56%	0.81%
Once-through	3.04%	1.18%	0.15%
Recirculating	0.45%	1.48%	0.39%

Alternative water sourcing and carbon mitigation technology. The PCR-GLOBWB 2 model provides runoff and sector-specific water withdrawal, including agricultural (irrigation and livestock), industrial, and municipal water withdrawal [42-45-47](#). We then estimate grid-level (0.5×0.5 degrees) water scarcity index (WSI) using Eq. 7 based on available total runoff and water withdrawal data for the historical period (1996-2005) for climate models (i.e., GFDL-ESM2M, HadGEM2-ES, IPSL-CM5A-LR) obtained from the ISIMIP online platform (ISIMIP 2b) (<https://www.isimip.org>). WSI is a widely used index to indicate the relative level of water scarcity. WSI equals 0.4 is usually used in prior studies as the threshold to indicate whether or not a region is exposed to high water scarcity⁴⁸, with a larger WSI indicating a higher water scarcity. WSI<0.1 indicates low water scarcity, 0.2>WSI>=0.1 indicates moderate water scarcity, and 0.4>=WSI>=0.2 indicates medium water scarcity. Water scarcity index reflects the relative share of water available that is used (e.g., withdrawal-to-availability resource ratio). In our study, we primarily focus on high water scarcity (WSI>0.4), with the same threshold (WSI=0.4) used for different regions and time periods to indicate the spatial and temporal heterogeneities in water

scarcity across the world and with changing climate. As not all freshwater can be used by human population (e.g., environmental flow needs), $WSI > 0.4$ is widely considered to be a reasonable, even though not definitive threshold value⁴⁸.

$$WSI = W/Q \quad (7)$$

W refers to PCR-GLOBWB 2 simulated total water withdrawal: adding agricultural, industry and domestic water withdrawal; Q refers to PCR-GLOBWB 2 simulated total runoff.

Increasing concerns on water scarcity are expected to facilitate further deployment of dry cooling techniques in the next decade or so before a dominating penetration of wind and solar PV becomes feasible. This, however, may lead to notable water-carbon tradeoffs due to much higher efficiency losses for dry cooling fleets, especially for thermal units in hot and arid regions. Therefore, we further integrate alternative water sources (i.e., wastewater and brine water) and carbon storage into power sector technology planning, such that to explore opportunities in resolving dry cooling-associated water-carbon tradeoffs.

We first integrate thermal generation units with grid-level global baseline water scarcity map to identify different generation units' exposure to local water scarcity. Globally, we assume dry cooling techniques can potentially be employed across the locations where thermal (i.e., here includes fossil and biomass) generation units use freshwater for cooling purposes and are exposed to high water scarcity ($WSI > 0.4$). Supposing dry cooling technique is employed, we then estimate the resulting unit-level efficiency loss and CO_2 emissions for supplying per unit (MWh) additional electricity. We emphasize that here we do not mean to predict future penetration rates of dry cooling techniques. Instead, we try to explore whether existing thermal generation units' locations would be suitable for utilizing dry cooling for alleviating water stress, considering the underlying water-carbon linkages. As PCR-GLOBWB 2 model does not include thermoelectric cooling water withdrawal for power generation, our estimated water scarcity index is conservative.

For generation units suffering from notable efficiency loss if dry cooling were utilized in water scarcity alleviation, we further explore the possibility of using alternative water sources to substitute dry cooling in mitigating freshwater dependence. Due to substantial treatment costs of wastewater and brackish water, we assume once-through cooling will be retrofitted to recirculating cooling when relying on alternative water for water scarcity alleviation. We obtain grid-level wastewater availability from Jones et al. (2021)²¹, which summarizes domestic and manufacturing total wastewater production (WWp, 359.4 bcm/yr), collection (WWc, 225.6 bcm/yr), treatment (WWt, 188.1 bcm/yr), and intentional reuse of treated wastewater (WWr, 40.7 bcm/yr) at 5 arc-minutes resolution. Notably, untreated wastewater can also be reused intentionally, while both treated and untreated wastewater can be reused unintentionally, which are not included in WWr²¹. Refer to Jones et al. (2021)²¹ for more details on global wastewater data. We consider a series of collection distances (e.g., 10 km, 25km, and 50 km) to capture the range of wastewater accessibility for generation units requiring alternative water sources. Supplementary Table 2 summarize extra

energy consumption and associated CO₂ emission intensities for wastewater treatment (Supplementary notes).

Additionally, we obtain CO₂ storage with commercial prospect together with brine water availability based on the CO₂ Storage Resource Catalogue (CSRC) Cycle 2, which assessed over 700 CO₂ Storage Resources sites, including both saline aquifers, as well as oil and gas fields⁴⁹. CSRC is a second update of the Oil and Gas Climate Initiative funded program aiming at gaining commercial readiness of geologic CO₂ storage resources across global key markets⁴⁹. Similarly, we assume a collection distance ranging from 10 to 25 and 50 km, with a series of brine extraction ratio (ER) of 20% (low), 50% (mid), and 90% (high)⁵⁰⁻⁵². Among locations with notable water-carbon tradeoffs if dry cooling were employed, alternative water sourcing can be utilized to substitute dry cooling if sufficient alternative water (e.g., wastewater and/or brine water) are available. We only consider brine extraction from candidate CO₂ storage sites. This is because brine extraction from CO₂ storage is considered a pressure management practice, which will inject CO₂ while extracting brine water, which can therefore prevent potential hazards such as land subsidence due to brine extraction. In addition, because brine extraction is a pressure management practice for CO₂ storage, the cost of brine extraction may be partially covered by geological carbon storage. However, if alternative wastewater and brine water are insufficient, dry cooling technique will then still be needed for freshwater mitigation while emitting additional CO₂ emissions. We hence further evaluate those units' nearby geologic CO₂ storage capacity to explore site-specific difficulty in tackling dry cooling-associated CO₂ penalty (e.g., increasing CO₂ emissions). In addition to currently-identified CO₂ storage capacity with commercial prospects, we also evaluate global geological potential of carbon storage and the associated brine water capacity based on the U.S. Geological Survey (<https://certmapper.cr.usgs.gov/data/apps/world-maps/>), which are much larger than those with commercial prospects. Supplementary Tables 2-3 summarize extra energy consumption, associated CO₂ emissions, and water demand intensities for brine water treatment and for conducting CCS. We further estimate the extra energy (and associated CO₂ emissions) and water for CCS deployment and alternative water (wastewater and brine water) treatment, which may (partly) dampen the role of alternative water and CCS in resolving dry-cooling associated water-carbon tradeoffs (Supplementary notes).

Data availability

Data used to perform this work can be found in the Supplementary Information. Numerical results for Figs. 1–5 and Extended Fig. 1 will be provided with this paper as source data, any further data that support the findings of this study are available from the corresponding authors upon request.

Code availability

Computer code or algorithm used to generate results that are reported in the paper and central to the main claims are available from the corresponding authors upon request.

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Author Contributions

Y. Q. and C. H. designed this study, Y. Q., W.Y., S. L., H. D., W. N. B. J., L. H., and C. H., analyzed the data, Y. Q., E. B., D. G., J. B., and G. H. wrote the paper with input from all co-authors.

Competing interests

The authors declare no competing financial interests.

Figure captions

Figure 1 | Age structure of global dry cooling generation capacity and associated water-carbon interactions. a,d, Curves indicate the estimated percentage of avoided water withdrawal (Δ Water withdrawal) and increased CO₂ emissions (Δ CO₂) from each age cohort to the sum of all ages' (a) non-fossil and (d) fossil dry cooling units in comparison to counterfactual cases of once-through freshwater cooling (dry VS. once). b,c, The operating capacity of non-fossil (b) and (c) fossil units where the youngest fleets are at the bottom. The dominance of young fleets indicates an apparent more recent dry cooling technology deployment. Here, generation units that began operating in 2015 are defined as 0 years old. Age-specific avoided water withdrawal and consumption, together with increased CO₂ emissions for dry cooling units in comparison to counterfactual cases of once-through and recirculating cooling are shown in Supplementary Fig. 2.

Figure 2 | Spatial pattern of global dry cooling generation units associated water-carbon interactions. Geospatial distribution of unit-level (a) avoided water withdrawal, (b) increases in CO₂ emissions, (c) increases in energy efficiency loss, and (d) increased CO₂ emissions per avoided water withdrawal for dry cooling units compared with once-through freshwater cooling. The geospatial heterogeneities in water withdrawal mitigation and carbon penalty is apparent. Map comparisons with recirculating cooling are shown in Supplementary Fig. 3. Global basemaps are based on Natural Earth⁵³ and plotted with python3.9.13.

Figure 3 | Share of generation capacity and water-carbon impacts by efficiency loss. In each panel, bars from left to right show the fraction of installed capacity (MW), electricity generation (ELE), avoided water withdrawal relating to once-through (AWW (Once)) and recirculating (AWW (Tower)) cooling, and increased CO₂ emissions relating to once-through (CO₂ (Once)) and recirculating (CO₂ (Tower)) cooling. Six vertical bars in each panel are composed by units with four levels of dry cooling units' efficiency loss. Panels are broken down by major fuel types, regions, and seasons. Biomass CO₂ emissions are shown to indicate potential carbon capture opportunities although they are considered close to carbon neutral. Numbers on top of each panel represent total electricity generation for specific fuel types, regions, and seasons.

Figure 4 | Increasing dry cooling units' efficiency loss under a warming climate. The temporal evolution of (a) ambient temperature (Tas), (b) efficiency loss, and (c) their relative increasing rates in comparison to history (1996-2005) (Δ Efficiency loss/ Δ Tas) under different climate warming scenarios (early century: 2020-2039, mid-century: 2040-2059, and late century: 2060-2099) for existing global dry cooling generation units weighted by generation capacity. As we use 10yr average GCM output, 2025 indicates the average value of the period of 2020-2029 and so on. In (a,b,c), colored shadings indicate multi-model ranges for each RCP scenario; colored vertical

lines represent the corresponding value ranges in 2090-2099, with the whole range, the triangles, and the short horizontal line represent the min-max, 25th and 75th, and mean values, respectively. d, e, f Spatial pattern of efficiency loss scaling factor in 2050-2059 relative to the beginning of the simulating period (2020-2029) to the Tas scaling factor ((Efficiency loss₂₀₅₀₋₂₀₅₉ /Efficiency loss₂₀₂₀₋₂₀₂₉) / (Tas₂₀₅₀₋₂₀₅₉/ Tas₂₀₂₀₋₂₀₂₉)) for each warming scenario. Global basemaps are based on Natural Earth⁵³ and plotted with python3.9.13. Unit-level efficiency loss responses to ambient temperature and scaling factors for 2090-2099 are shown in Extended Fig. 1 and Supplementary Fig. 7.

Figure 5 | Historical integrated dry cooling with alternative water sourcing and carbon mitigation. (a) Unit-level efficiency loss for thermal generation units if equipped with dry cooling exposed to historical high water scarcity (WSI > 0.4) (1996-2005). (b-e) Ratio of grid-level total cooling Water withdrawal to Alternative water sources (wastewater and brine water) (**RWA**) for magenta high-efficiency-loss units if equipped with dry cooling in Fig. 5a. Alternative water sources refer to: (b) the sum of intentional reusable wastewater and potentially commercial-ready brine water with a collection distance of 25 km (WWr+C-Brine), (c, d) the sum of total produced wastewater and C-Brine (WWp+C-Brine) with a collection distance of 25 km and 50 km, and (e) the sum of intentional reusable wastewater and brine water based on the USGS geological survey (WWr+USGS-Brine) with a collection distance of 25 km. Brine water extraction rates in b-e are all 50%. (f) Accessibility to nearby USGS carbon storage for generation units lacking sufficient alternative water (magenta dots in e). (g-j) Pie charts show the relative share of generation capacity falling into different levels of RWA, varying by (g) alternative water (WWr+C-Brine) collection distance (10 to 50 km), (h) the portion (WWr, WWt, WWc) of total wastewater produced (WWp), (i) brine water extraction rates (ER) (20% to 90%), and (j) brine water capacity (C-Brine to USGS-Brine) and collection distance (10 to 50 km). Numbers in g-j indicate the proportion of operating generation capacity belonging to the lowest ($\leq 25\%$, blue) and highest ($>100\%$, magenta) RWA level. $RWA \leq 100\%$ indicate alternative water is sufficient to substitute dry cooling for freshwater mitigation.

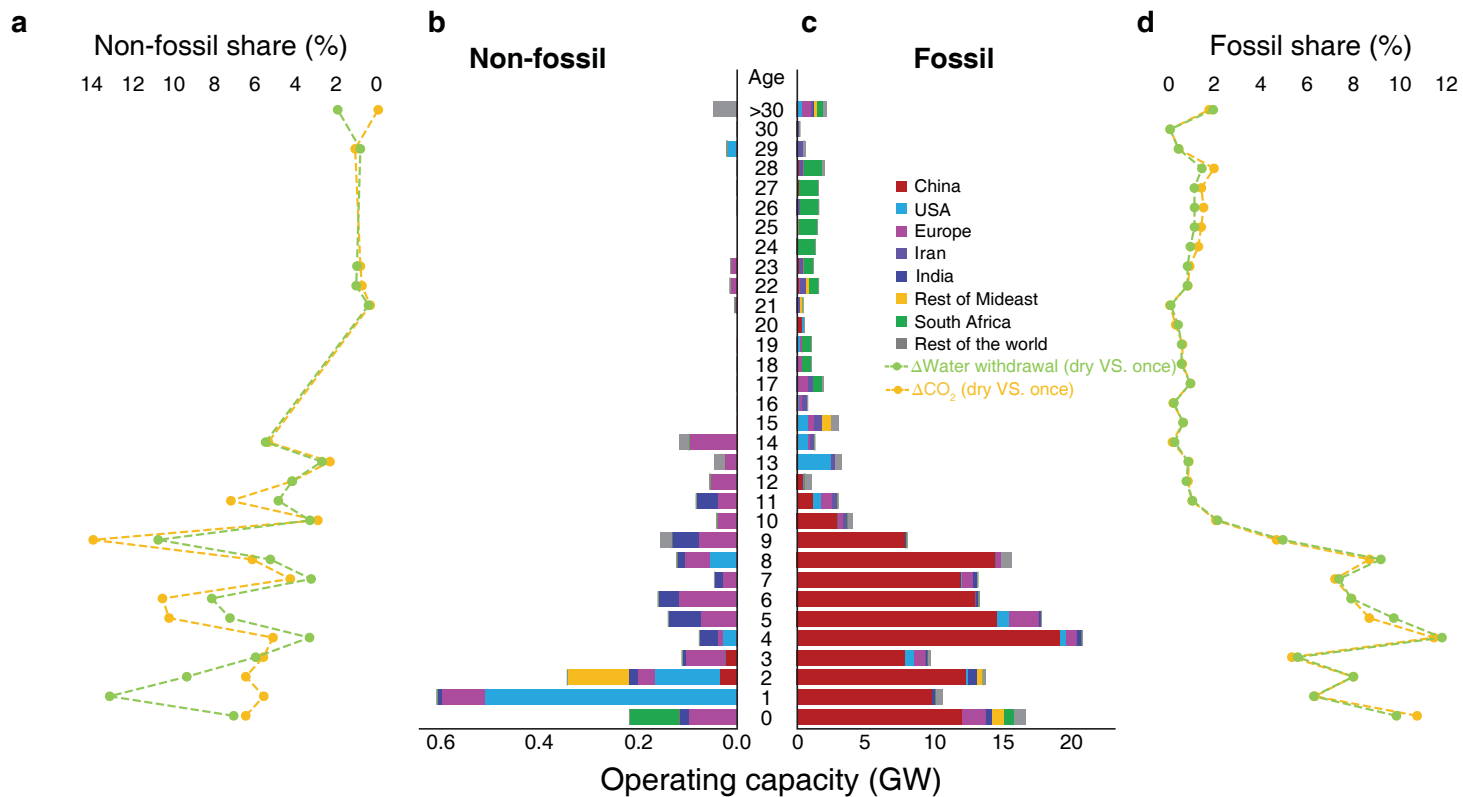
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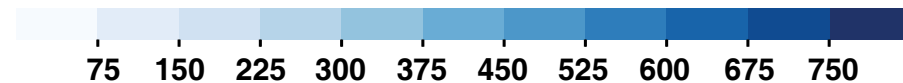
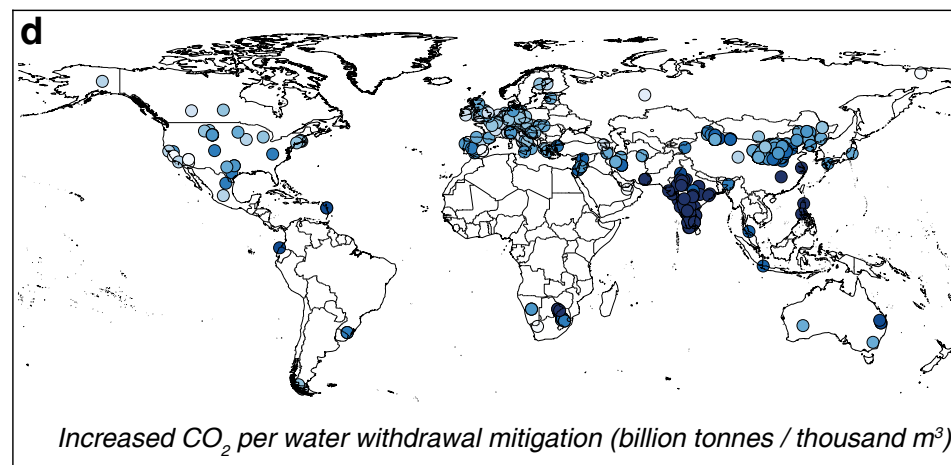
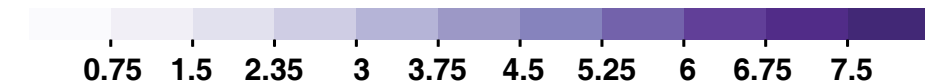
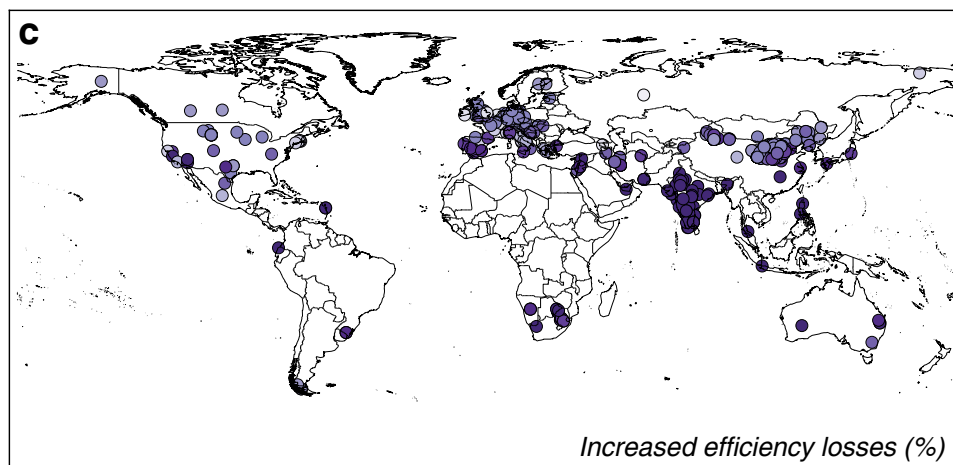
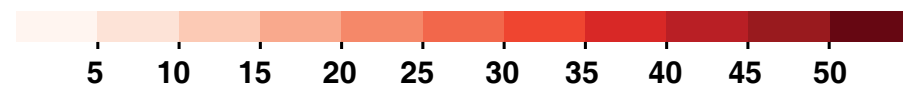
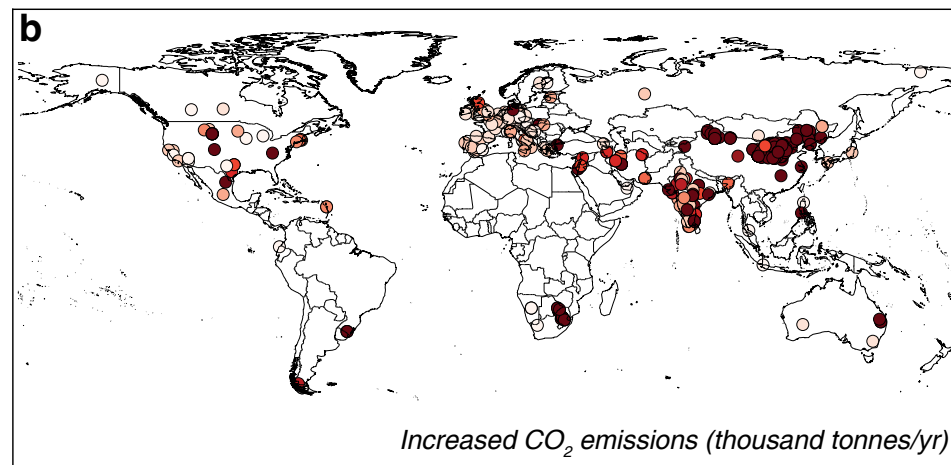
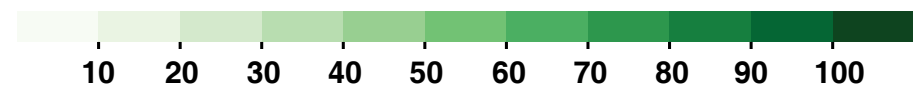
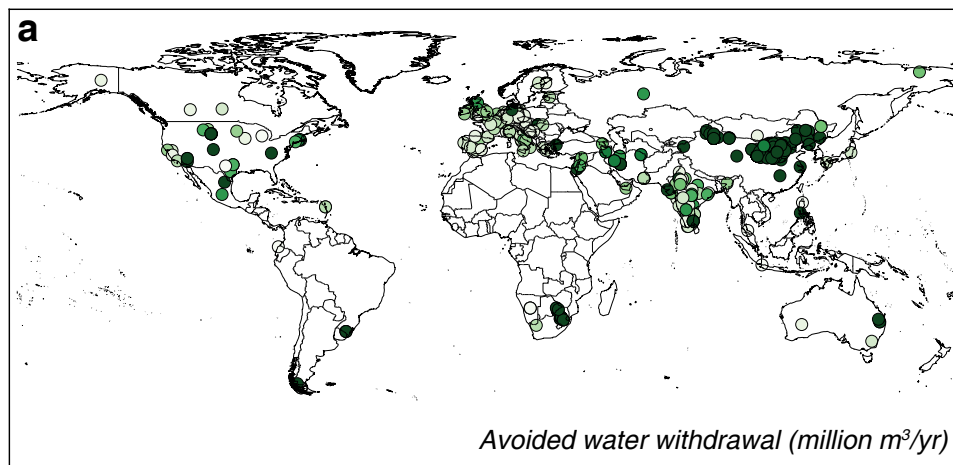
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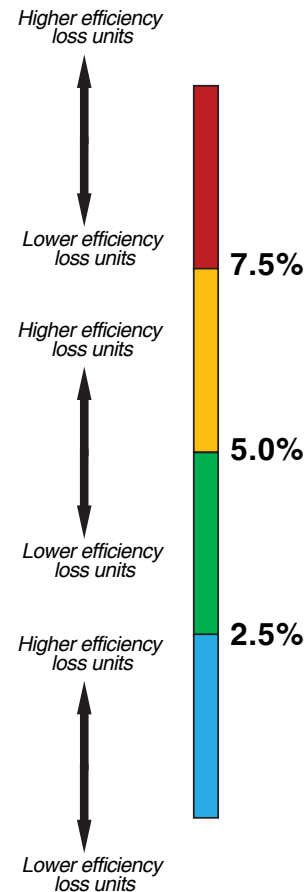
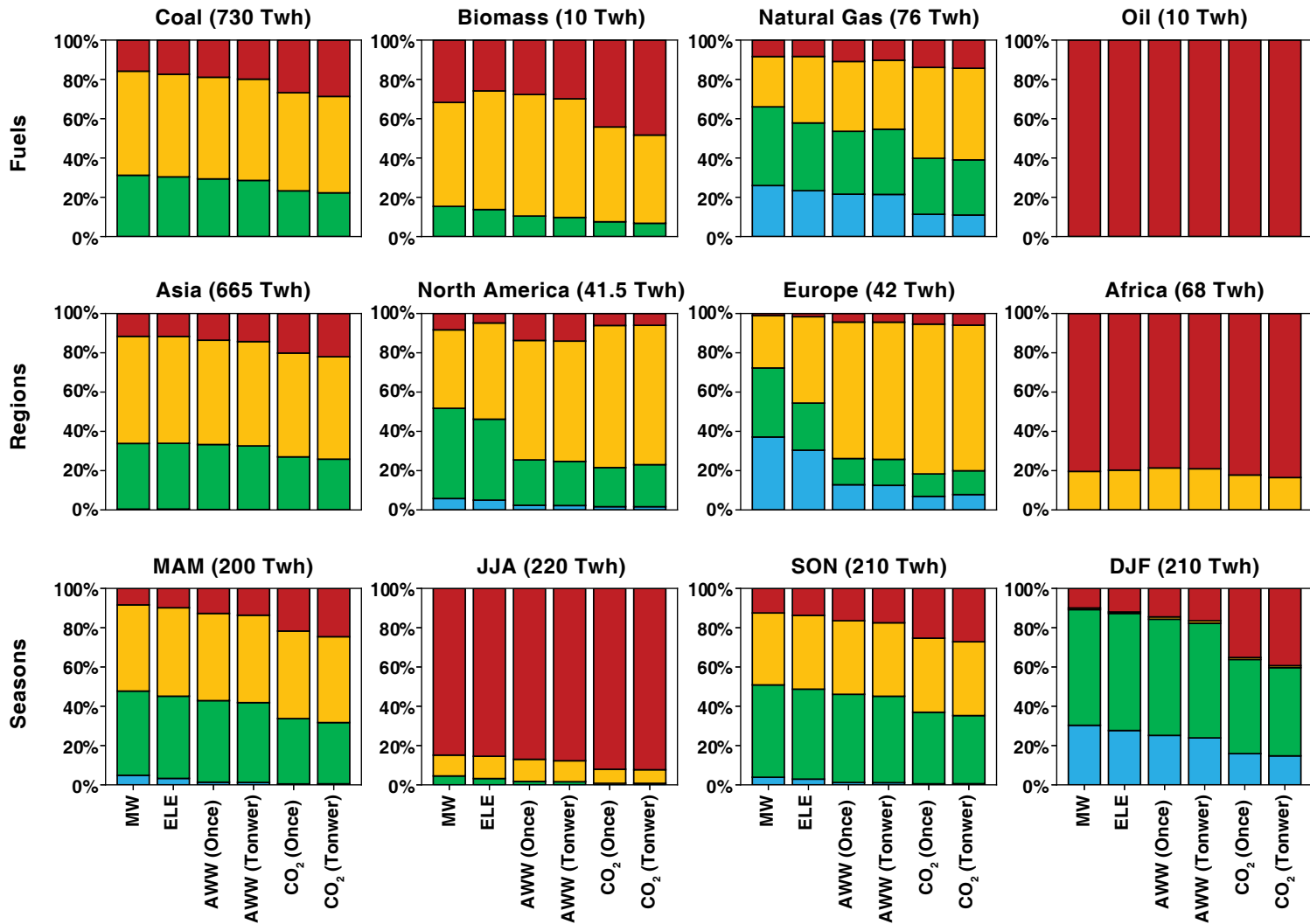
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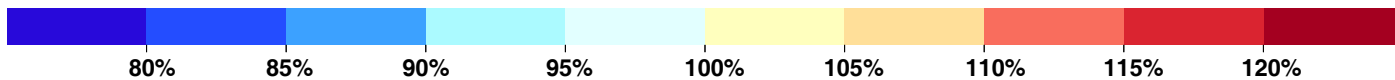
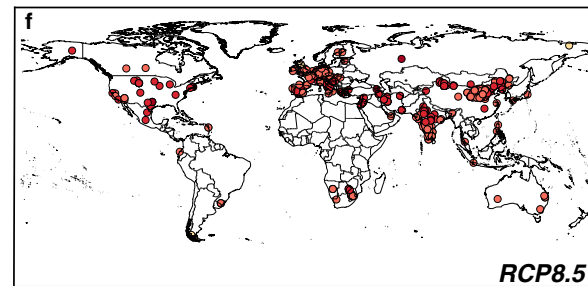
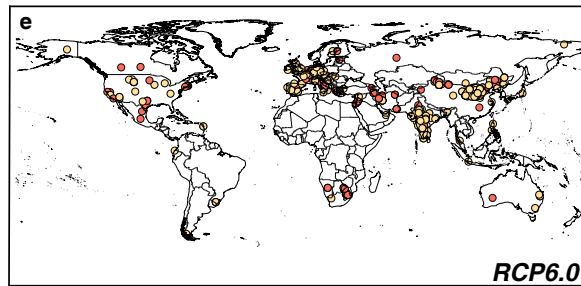
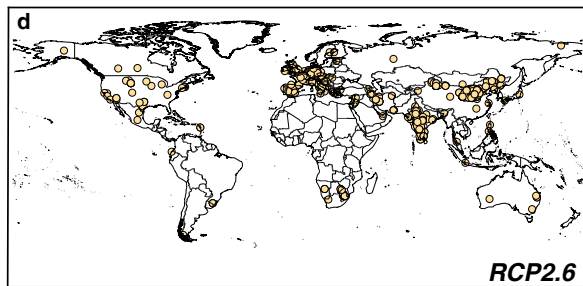
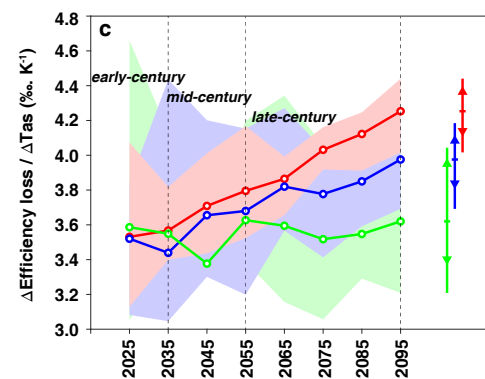
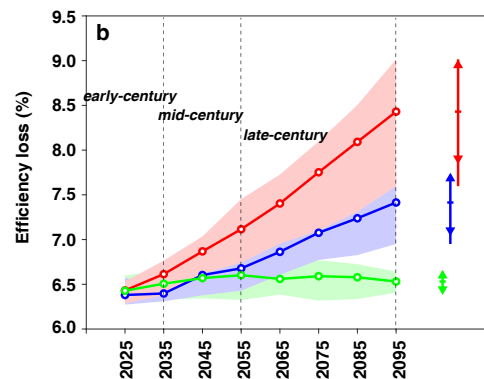
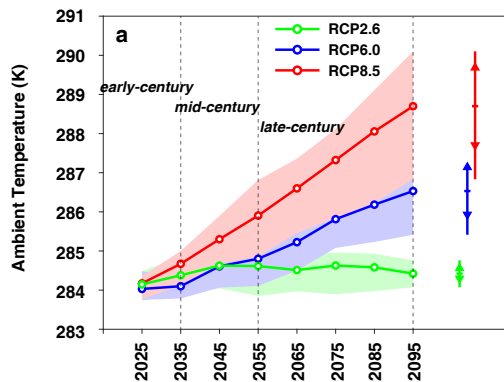
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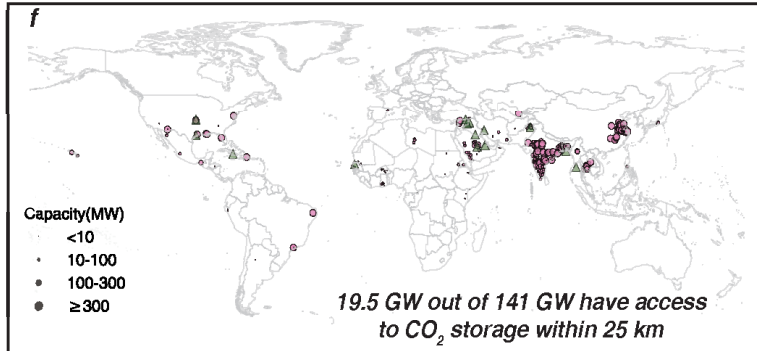
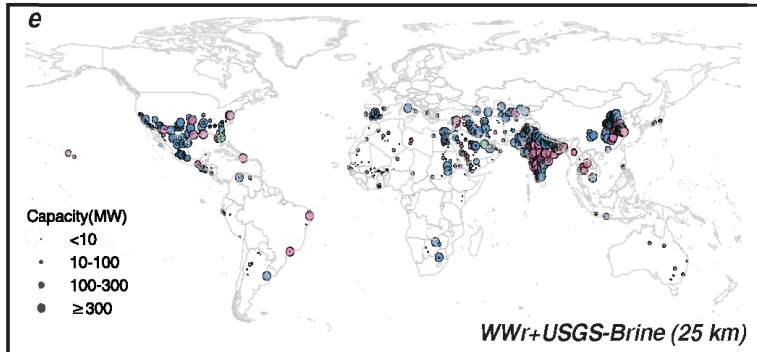
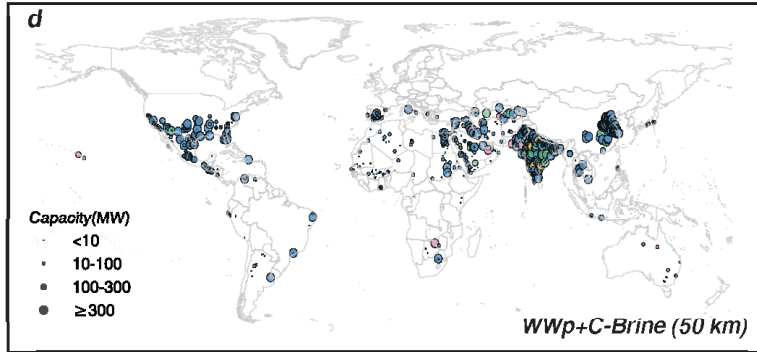
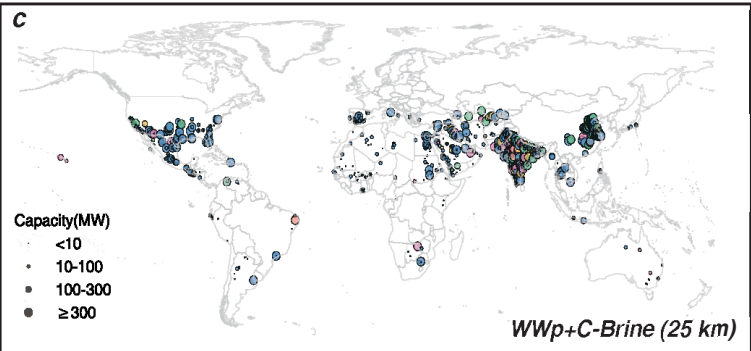
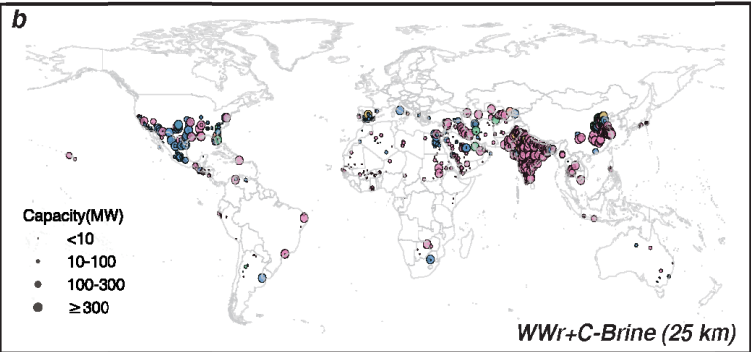
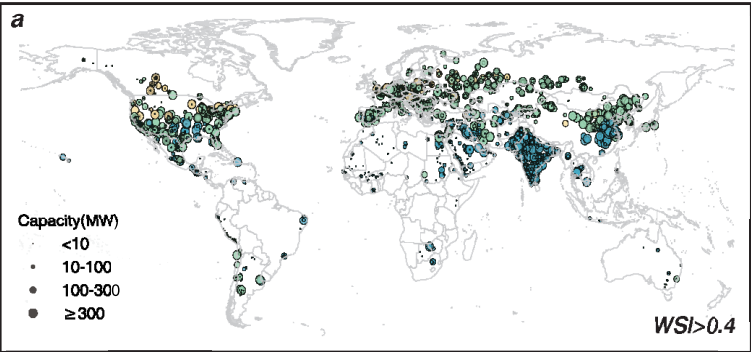
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● Units with insufficient nearby alternative water
▲ CCS of USGS geological survey

